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Dual-Programmable Architected Magnetic Soft Materials: Tuning Mechano–Electric Responses by Inverse Design

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ABSTRACT

Materials that can deform, sense, and autonomously generate power in response to wireless magnetic fields, through both magnetic-actuated shape transformation and charge generation, are an emerging focus in advanced functional materials research. Magneto–mechano–electric intelligent materials, created by embedding ferromagnetic particles into soft elastomers with immobile charges, are bringing this vision closer to reality. Under wireless magnetic actuation, they undergo rapid deformation that drives charge flow, enabling controlled responses. Yet realizing such materials to unleash their full potential remains a major challenge because of the complex multiphysics coupling under large deformation and independent mechanical and electrical responses. In this work, we create a new class of dual-programmable magnetic soft materials with unprecedented dual properties by establishing a multiphysics inverse-design framework that simultaneously optimizes geometry and remanent magnetization distributions to tune both magneto–elastic and magneto–electric behaviors. The framework is complemented by a tailored fabrication approach and experimental validation, forming a comprehensive and automated pipeline for creating multifunctional advanced materials for biomedical devices with dually programmable behaviors. Its effectiveness is demonstrated through an electrical stimulation guidewire for endovascular therapy, mechano–electric treatment robots for injury-specific tissue recovery, and multifunctional self-powering and self-sensing bio-robots with on-demand mechanical motions. Together, these results establish an effective inverse-design-to-fabrication paradigm for dual-programmable magnetic soft materials in next-generation biomedical applications.

1 | Introduction

Imagine a soft material that can undergo finite deformation in a programmable manner under a magnetic field, while simultaneously generating programmable electricity. Such a material holds great promise for developing biomedical devices. It can be actuated wirelessly, contactlessly, and rapidly, making it well-suited for functioning in confined biological environments using minimally invasive surgery [1, 2]. The mechanical compliance and softness enable it to conform to and integrate with biological tissues [3]. At the same time, the material offers dual outputs with full programmability, including mechanical and electric stimulation, that are highly useful for therapeutic intervention [4] as well as energy harvesting and self-sensing functionalities in the tissue environment.

These materials are rarely found as a single phase in nature, because at the material level, the presence of magnetism usually

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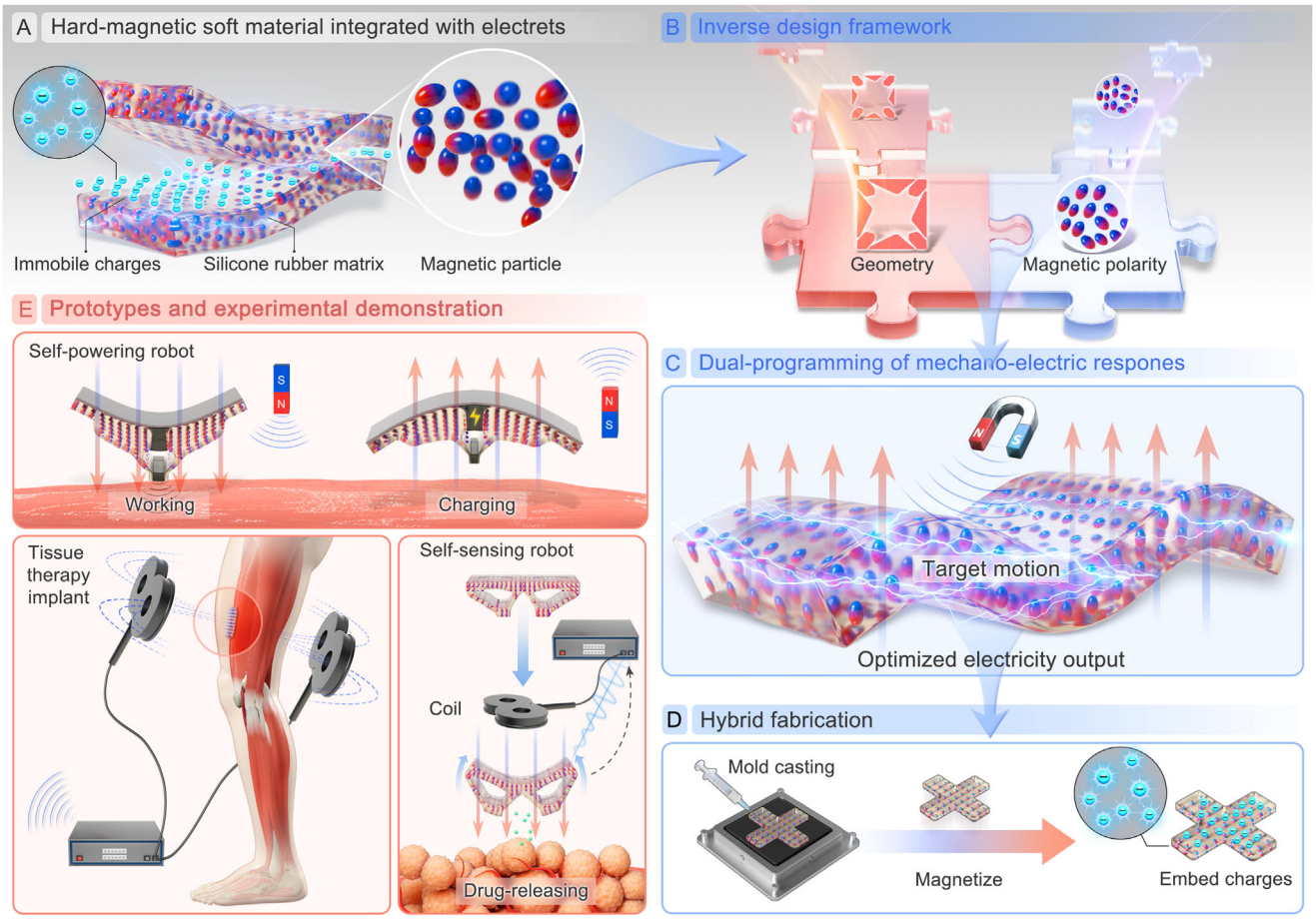


FIGURE 1 | Pipeline for realizing dual-programmable architected magnetic soft materials: (A) Composition of hard-magnetic soft materials integrated with electrets; (B) Inverse design framework that simultaneously optimizes geometry and magnetic polarity; (C) Dual-programming of magneto-elastic deformation and magneto–electric generation; (D) Hybrid fabrication via mold-casting and charge embedding; (E) Prototypes and experimental demonstration of dual-programmable magneto–electric materials including tissue therapy implants and multi-functional robots.

prevents them from being electric [5]. An emerging type of soft magneto–electric composite material offers a promising solution for creating magneto–mechano–electric soft matter. The composite embeds pre-magnetized ferromagnetic particles and immobile charges into soft elastomers, termed as hard-magnetic soft electrets [6, 7], as shown in Figure 1A. The immobile charges can be embedded into the composite through the incorporation of charge-storing carriers with stable charge-attachment property, such as polytetrafluoroethylene (PTFE) film [6] or silica nanoparticles [8]. The coupling mechanism leverages mechanical deformation as a physical bridge between magnetic actuation and electric output. When the material is exposed to a magnetic field, the embedded magnetic particles reorient, driving deformation of the composite. The mechanical deformation changes the electric field generated from the immobile charges, thereby inducing an electric response.

Translating magnetic soft materials into real-world applications requires addressing a key design challenge. The various material components should be carefully engineered so that both the mechanical and electrical outputs are effective and, ideally, programmable to meet specific biomedical needs, which often vary across different practices and patients. Intuition-based methods [6, 7] have demonstrated promising designs. However, unleashing the full potential of electricity generation and arbitrary yet precise deformation control remains underdeveloped. Addressing this task is challenging because of the highly complex coupling among the three physics (i.e., magneto–mechano–electric) and the vast multi-variable design space (i.e., magnetic polarity patterns and material geometry at each location). To address the design challenge, computationally aided design [9, 10] offers a powerful avenue for tackling this complex design problem. While prior studies have explored design methods for magnetic soft materials using gradient-based optimization methods [11–14], non-gradient design strategies [15, 16], and data-driven models [17, 18], few efforts have focused on architecting electret-integrated magnetic soft materials to enable dual mechanical and electrical programmability.

Here, we address this critical gap by creating a dual-programmable magnetic soft material with embedded electrets by introducing an automated, multiphysics, and gradient-based inverse-design framework that simultaneously optimizes material geometry and the spatial distribution of remanent magnetization (indicating magnetic polarity) with magneto–mechano–electrical coupling, as illustrated in Figure 1B. The optimization framework features a multi-objective function that optimizes both charge output and mechanical deformation, as

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well as several function-based constraints. Driven by analytical gradient and physics-based simulation, this automated generative framework enables precise programmability of both magneto-mechanical and magneto–electric responses (Figure 1C). Details are included in Section 4 and Supporting Information. Through tailored fabrication and experimental validation (Figure 1D), we demonstrate dual-programmable architected materials that can address diverse biomedical applications (Figure 1E). They include (1) a magnetic guidewire working in blood vessels for medical treatment with maximized electrical output, (2) therapy robots capable of delivering precise injury-specific mechanical and electrical treatments, and (3) Energy-harvesting and self-sensing motion-control bio-devices that can detect mechanical motion, wirelessly self-power microelectronics (e.g., an LED), while enabling on-demand mechanical actuation (e.g., drug release). Collectively, these advances establish an inverse design and hybrid fabrication paradigm for architecting dual-programmable magnetic soft materials, paving the way for next-generation intelligent multifunctional biomedical robots and implants.

The novelty of this work lies in introducing electricity as a third coupled physical field into the topology optimization of hard-magnetic soft materials, enabling the simultaneous inverse design of mechanical deformation and electrical output in magneto–mechano–electric systems. Prior studies have demonstrated electret-embedded hard-magnetic soft materials through numerical modeling and benchmark cantilever experiments [6, 7], while topology optimization of hard-magnetic soft materials has primarily focused on programming magneto–mechanical deformation and motion [11]. In contrast, this inverse design work explicitly solves the electric field strongly coupled with the nonlinear magneto-mechanical system, allowing the electrical output to be directly optimized together with the mechanical behaviors. The optimized designs are physically realized and validated by systematic fabrication, characterization, and experiments under large deformations.

2 | Results

2.1 | Magnetic Guidewire with Maximized Electrical Stimulation

Magneto–electric materials enable wireless electrical stimulation from within blood vessels, with potential applications in various treatments such as neuromodulation [19] and antithrombotic therapy [20]. The magnetic guidewire example is illustrated in Figure 2A. This approach offers a unique advantage by accessing hard-to-reach regions through the blood vessel system [19]. In this section, we aim to advance the technology using a dual-programmable architected magnetic soft guidewire that enables maximized electrical output and correspondingly maximized bending (leading to a large workspace [21]). In Figure 2B, we formulate a simplified guidewire design problem. The design space is defined with a long rectangular domain (22mm × 2.2mm), where we optimize the remanent magnetization distribution. We note that the out-of-plane thickness is 12 mm, selected for fabrication feasibility and experimental accuracy, as smaller samples would produce weaker signals that are more susceptible to environmental noise. The design remains scalable within the proposed optimization framework. The domain is fixed at one end, with electrodes applied to the left and right sides under short-circuit conditions. A layer of negative embedded charge is distributed along the midline of the domain. To illustrate the deformation-mediated magneto–electric behavior, first consider the system in the absence of a magnetic field, as shown on the left of Figure 2B. The two grounded electrodes of the guidewire hold positive free charges, which are balanced by the embedded negative charges. Upon applying a magnetic field, the guidewire undergoes magnetically actuated bending, and this deformation alters the electric field distribution, causing the free charges to redistribute across the two short-circuited electrodes and generate an electric current, as shown on the right of Figure 2B.

The optimization setup and results are presented in Figure 2C. We consider eight candidate orientations of remanent magnetization along with a non-magnetized option. An external magnetic field is applied from right to left. The optimization objective is to maximize the output charge (ΔQ, defined as the variation of induced positive charges before and after magnetic actuation) on the left electrode. Based on previous findings for similar slender-beam setups, a larger output charge corresponds to greater bending [7]; hence, maximizing the output charge inherently achieves dual programmability in both electrical and mechanical responses. Using the developed inverse-design framework, we generate an optimized design (Design 1-2). For comparison, we also include an intuitive design (Design 1-1) with purely downward magnetization. The rationale behind the intuitive design arises from the observation that the magnetic torque driving the initial bending (without considering geometric nonlinearity) reaches its maximum when the remanent magnetization is oriented perpendicular to the external magnetic field [11, 22].

Figure 2D compares the performance of Designs 1-1 and 1-2 in terms of ΔU and ΔQ under an external magnetic field ($|B_a|$) ranging from 0 to 20 mT. Design 1-1 exhibits a larger ΔU than Design 1-2 when $|B_a| < 6.5$ mT, because its intuitive magnetization distribution leads to a larger initial bending torque. In contrast, Design 1-2 outperforms Design 1-1 for $|B_a| > 6.5$ mT, as the optimization accounts for geometric nonlinearity in magnetic actuation. This enhanced deformation allows Design 1-2 to achieve an 88% increase in ΔQ at $|B_a| = 20$ mT, as confirmed by finite element analysis (FEA). The consistent relationship between ΔQ and $|B_a|$, as well as ΔU and $|B_a|$, further demonstrates that large bending deformation is the dominant magneto–electric mechanism in this cantilever beam setup, in agreement with the findings of [7].

Experimentally, we fabricate and test the two designs. As illustrated in Figure 2E, the charges are embedded into the magnetic soft material through PTFE layers serving as charge carriers, owing to PTFE's strong tribonegative properties and excellent ability to trap negative charges. PTFE films are triboelectrically charged and subsequently sandwiched between magnetic elastomers. Two error indices are defined to quantify the discrepancies in the actuated deformation and the initial centerline deflection:

$$e_{\Delta U} = \frac{\left|\Delta U_{\mathrm{FEA}} - \Delta U_{\mathrm{Exp}}\right|}{\left|\Delta U_{\mathrm{Exp}}\right|}, \quad e_p = \frac{\left|\bar{u}_{x,\mathrm{Exp}} - \bar{u}_{x,\mathrm{FEA}}\right|}{L_y} \tag{1}$$

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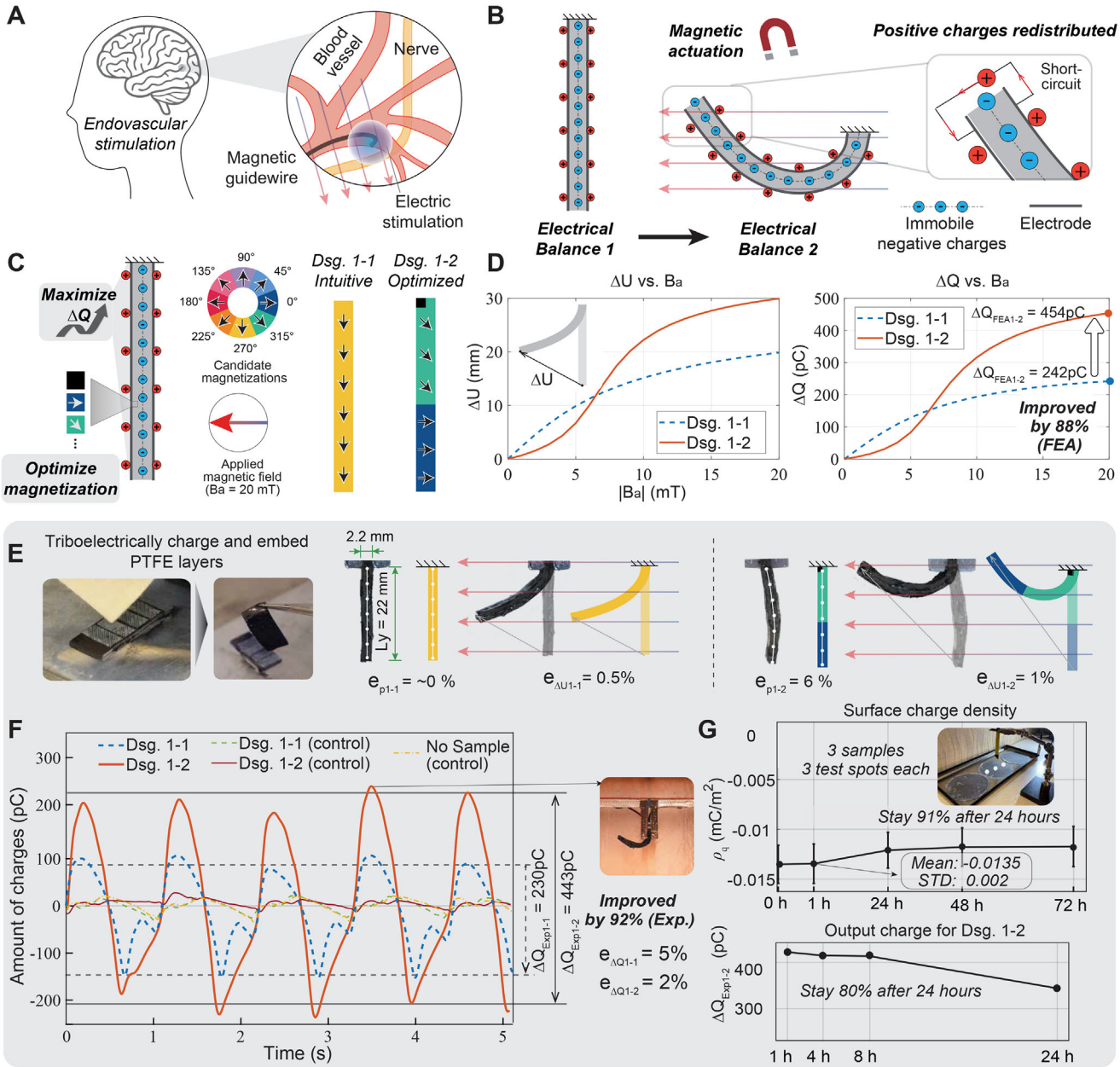


FIGURE 2 | Magnetic guidewire with electric stimulation: (A) Schematic showing electrical stimulation delivered by a magnetic guidewire in blood vessel; (B) Working principle of electricity generation induced by magnetically driven deformation; (C) Design domain and setup comparing an intuitive magnetization pattern (Design 1-1) with an optimized one (Design 1-2); (D) Simulations of tip displacement and output charge under different magnetic field strengths; (E) Triboelectric charging process; (F) Comparison of experimental and simulated static deformations for both designs; (G) Experimentally measured output charge showing the enhanced performance of the optimized design; (H) Surface charge characterization for charged PTFE on magnetic elastomer (sample size $n = 3$) and output charge testing for Design 1-2.

where ΔU denotes the tip displacement before and after magnetic actuation, $\bar{u}_x$ is the average displacement in the horizontal x direction along the centerline of the magnetic guidewire, and L_y is the guidewire length used for normalization. For Design 1-1, the FEA prediction agrees well with the experimental measurements, with relative errors of $e_{\Delta U1\text{-}1} = 0.5\%$ and $e_{p1\text{-}1} \approx 0\%$. For Design 1-2 with optimized magnetization, an initial deformation is observed at $|B_a| = 0$ due to dipole–dipole interactions. The corresponding errors are $e_{\Delta U1\text{-}2} = 1\%$ and $e_{p1\text{-}2} = 6\%$. The discrepancies are partially introduced by the modeling limitations of the decoupled magneto–mechanical model used in the current study. We note that strongly coupled models (e.g., [23]) can further reduce the prediction error, with more detailed investigation provided in Section S4.

Furthermore, we measure the output charge, ΔQ, for both designs, as shown in Figure 2F and Movie S1. The test is conducted at an actuation frequency of 0.9 Hz, with a sinusoidal magnetic field applied. From our measured data for Designs 1-1 and 1-2, we observe nearly sinusoidal charge signals with consistent peaks and valleys. A slight kink is observed in the test sample of Design 1-1, which may result from dynamic

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oscillations during the experiment. The difference between peak and valley values represents the relative charge amount, yielding $\Delta Q_{\text{Exp1}-1} = 230\,\text{pC}$ and $\Delta Q_{\text{Exp1}-2} = 443\,\text{pC}$, respectively, with a 92% improvement. The experimentally measured values and the observed improvement closely match the FEA results, $\Delta Q_{\text{FEA1}-1} = 242\,\text{pC}$ and $\Delta Q_{\text{FEA1}-2} = 454\,\text{pC}$. The relative errors between the numerical simulations and experiments for the two designs are $e_{\Delta\text{Q1}-1} = 5\%$ and $e_{\Delta\text{Q1}-2} = 2\%$, respectively.

To further evaluate the credibility of the measured data, we test control samples for both designs without embedded negative charges, as well as a control case with no sample (Figure 2F). We observe that these control cases yield much smaller electrical signals, with a magnitude of approximately 30 pC. We attribute this to electromagnetic interference from the environment in the lab, especially from the coil that applies the magnetic field.

To evaluate repeatability, multiple measurements of the surface charge density (ρ_q) and the output charge ($\Delta Q_{\text{Exp1-2}}$) for Designs 1–2 are conducted over time under a humidity of 30% (Figure 2H; refer to Section 4.8 for more characterization details). Both ρ_q and $\Delta Q_{\text{Exp1-2}}$ exhibit a gradual decrease, which can be attributed to charge perturbation on the PTFE layers. Nevertheless, 91% of the initial surface charge density and 80% of the initial output charge are retained, indicating good charge stability. The relatively greater degradation observed in the output charge is likely due to increased charge perturbation induced by continuous cyclic deformation.

2.2 | Mechano–Electric Therapy Robots with Dual-Programmable Injury-Specific Mechanical and Electrical Treatments

Mechanical and electrical stimulation have been shown to promote tissue healing [4, 24, 25]. Here, we present a new class of mechano–electric therapy robots (Figure 3A) designed to deliver target stimulation to soft tissues by tuning dual-programmable deformation and electricity generation. These soft robots can be actuated remotely by magnetic fields, providing a wireless strategy to accelerate deep tissue repair.

The therapy robots feature a multilayer soft composite architecture (Figure 3B). The top layer serves as the design domain, beneath which upper and lower substrate layers sandwich a charge-carrier layer containing immobile negative charges. A bottom electrode is integrated to induce responsive positive charges and maintain direct contact with the tissue. For modeling, the robot is assumed to be anchored at the four corners of the tissue. In this configuration, the mechanical resistance of the tissue is neglected, and its bottom layer is treated as electrically grounded, allowing charges to flow freely to preserve charge balance. The design setup reflects several practical surgical considerations: (1) a sufficiently large contact and conductive area between the robot and tissue enhances the robustness of charge output; (2) a four-corner fixation setup is compatible with common surgical attachment methods, such as suturing or adhesive bonding; and (3) the magnetic members that could produce self-interaction on the top layer are firmly bonded to the upper substrate, ensuring structural integrity and facilitating surgical handling.

For the inverse design problem, we simultaneously optimize the geometry of the top layer and the in-plane remanent magnetization distribution, selected from a discrete set of eight candidate directions. The applied magnetic field is out-of-plane with $B_{a,z} = -20$ mT. We impose a volume constraint with an upper bound of 0.3. The optimization is formulated with two objectives to realize dual programmability (Figure 3C): (1) maximizing the out-of-plane displacement ΔU at prescribed control degrees of freedom (DOFs), corresponding to mechanical stimulation, and (2) maximizing the output charge ΔQ collected at the bottom electrode, enabling electrical stimulation.

To demonstrate the dual programming of mechanical deformation and electric stimulation, we compare two optimized designs in Figure 3C. Design 2-1 is optimized solely for maximum central deformation, reaching a peak displacement of $\Delta U_{\text{FEA2-1}} = 6.8$ mm and producing a charge output of $\Delta Q_{\text{FEA2-1}} = 215$ pC. In contrast, Design 2-2 is optimized to jointly maximize displacement and charge generation, leading to a distinct geometry and magnetization pattern. This design achieves an 82% higher charge output of $\Delta Q_{\text{FEA2-2}} = 392$ pC, at the cost of a 20% reduction in displacement to $\Delta U_{\text{FEA2-2}} = 5.64$ mm. Examining the magnetization orientations of both designs, we observe that the directions tend to point toward the control DOFs, which promotes out-of-plane deformation, consistent with the observations reported in [27]. Additionally, the geometry of the magnetized domains differs between Designs 2-1 and 2-2, reflecting the optimization's role in identifying deformation modes that more effectively generate charge. The inverse-design process is also presented in Movie S2.

To showcase the flexibility of the proposed inverse-design framework for case-specific medical scenarios, we prescribe an injury site with an irregular shape, as indicated in Figure 3D. The objective is to concentrate mechanical stimulation over the injury region while ensuring that electrical stimulation is maximized through the bottom electrode. To achieve this, four control DOFs following the injury pattern are assigned, and the optimization seeks to dually program the displacement at these DOFs as well as the charge output at the bottom electrode. The resulting design (Design 2-3) delivers spatially focused mechanical stimulation aligned with the injury pattern, achieving a localized average deformation of $\Delta U_{\text{FEA2-3}} = 4.59$ mm and an output charge of $\Delta Q_{\text{FEA2-3}} = 319$ pC. The displacement contour of ΔU_z (bottom right in Figure 3D) closely matches the geometry of the injured region and effectively covers the assigned control DOFs. This design highlights the capability of the optimization framework to generate injury-specific and dual-programmable mechano–electric stimulation patterns for targeted tissue therapy, offering strong potential for developing more adaptable and patient-specific therapy robots.

To validate our computational design, we fabricate Design 2-2 and conduct magnetic actuation deformation tests and output charge measurements (Figure 3E and Movie S2). The experimentally measured out-of-plane deformation at the central control DOF, measured using a depth caliper, is approximately $\Delta U_{\text{Exp2-2}} = 5.7$ mm, closely matching the FEA prediction and validating the accuracy of the simulation $\Delta U_{\text{FEA2-2}} = 5.64$ mm, with a relative error of $e_{\Delta\text{U2}-2} = 1\%$. Under dynamic magnetic actuation (0.9 Hz sinusoidal B_a varying from 0 to 20 mT), the device generated an average cyclic output charge of $\Delta Q_{\text{Exp2-2}} = 432$ pC, which

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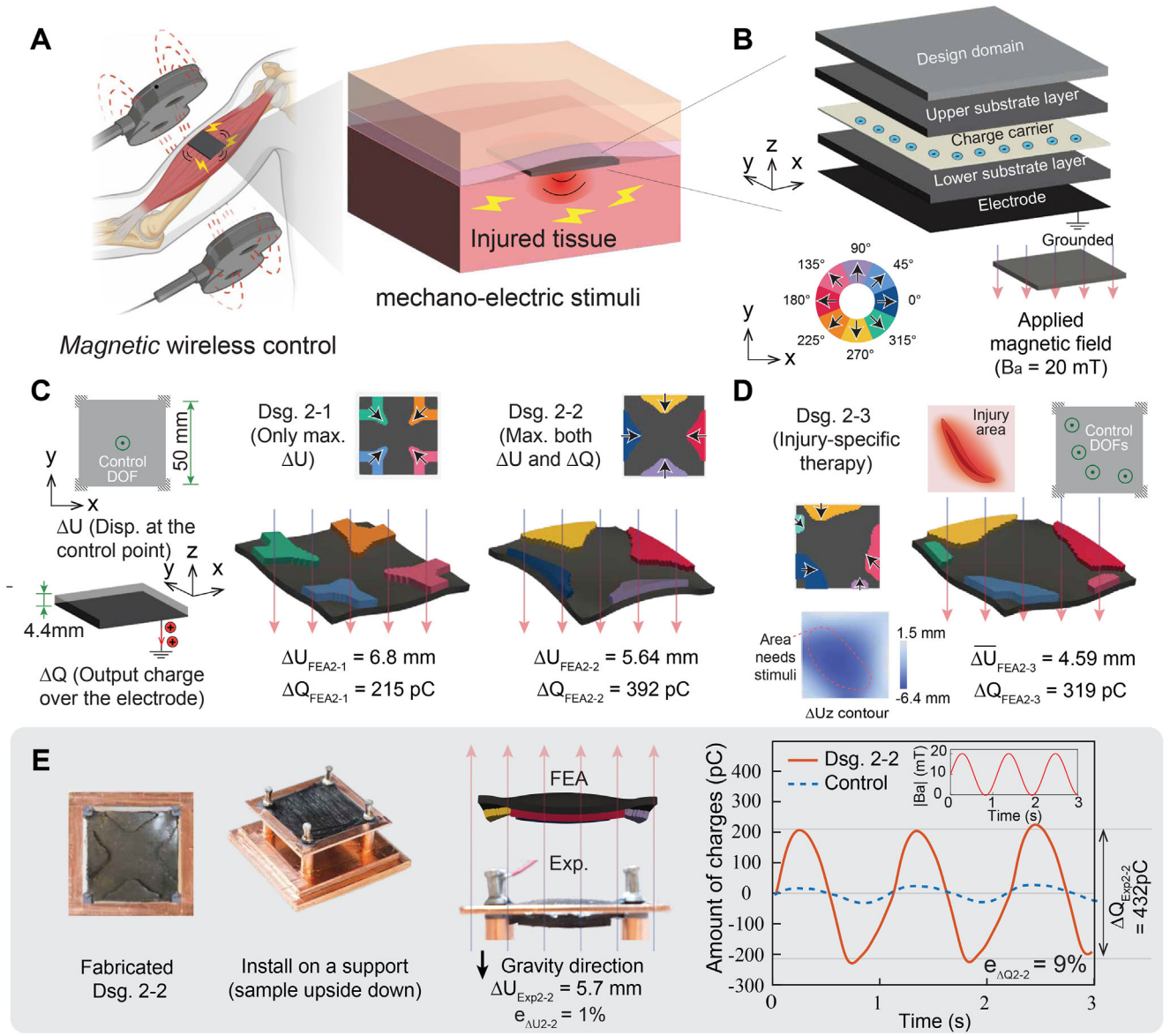


FIGURE 3 | Mechano–electric therapy robots: (A) Concept of magnetically actuated therapy robots delivering stimulation to accelerate tissue healing (adapted from [26]); (B) Design setup featuring a multilayer configuration; (C) Topology-optimized designs: Design 2-1 (maximizing displacement), Design 2-2 (jointly maximizing displacement and output charge), and Design 2-3 (targeted injury-specific stimulation); (D) Fabrication and experimental validation for displacement and output charge of Design 2-2.

is in good agreement with the quasi-static FEA prediction of $\Delta Q_{FEA2\text{-}2} = 392$ pC, with a relative error of $e_{\Delta Q2-2} = 9\%$. For comparison, the control sample without an embedded charge carrier produced much lower output, experimentally confirming that the charge observed in Design 2-2 results from the deformation-coupled magneto–electric effect.

2.3 | Energy Harvesting and Self-Sensing Motion-Control Multifunctional Bio-Robots

Another important application of this magneto–mechano–electric material lies in wireless energy harvesting and self-sensing. With its dual-programmable mechano–electric response, it enables advanced materials for multifunctional robots that are promising for biomedical use [28, 29]. In this context, we apply the proposed inverse-design framework to demonstrate two envisioned magneto-actuated bio-robots: a self-powered sensing robot as well as a self-sensing drug-releasing robot (Figure 4A,C, respectively). Building on concepts of enhancing magneto–electric effects through snapping mechanisms [30, 31] and recent advances in controllable snap-through driven by magnetic actuation [32–34], these devices exploit magnetically induced snap-through deformation to achieve rapid shape morphing and enhanced electric current flow.

Inspired by the design concept in [31], the design domain (Figure 4B) consists of an air gap between a fixed electrode and a compliant electrode bonded to the magnetic material. The charge-carrying layer is positioned directly on the fixed electrode, rather than being embedded within the magnetic material. This configuration enables the electret to achieve a

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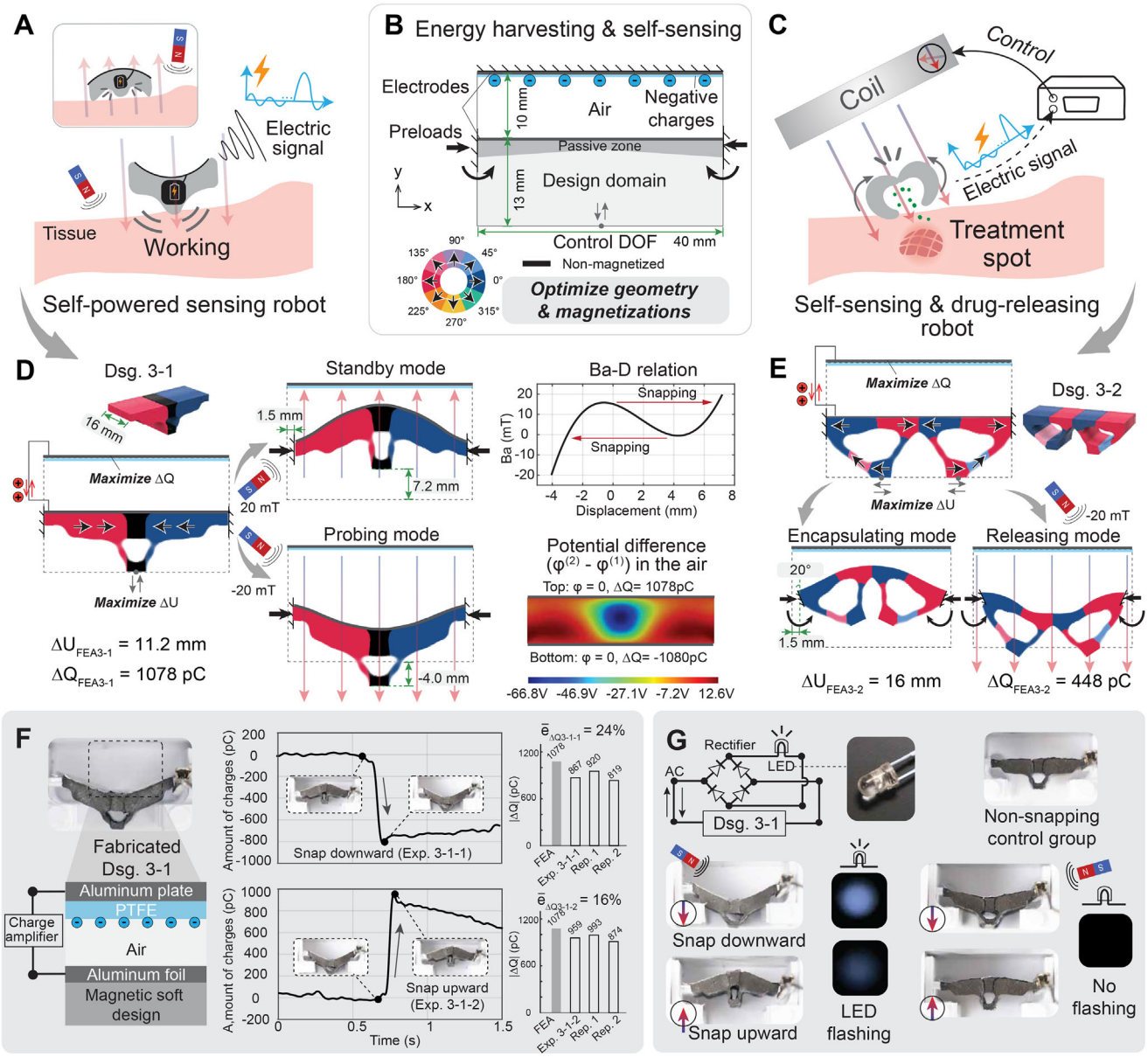


FIGURE 4 | Energy-harvesting functional bio-robots: (A) Concept of a self-powered sensing robot; (B) Design setup; (C) Concept of a self-sensing drug-releasing robot; (D) Optimized Design 3-1 enabling magneto-actuated self-powered sensing robot; (E) Optimized Design 3-2 enabling encapsulation without a magnetic field and magneto-actuated drug release; (F) Experimental measurement of the output charge (sample size $n = 3$) and demonstration of the self-sensing capability for Design 3-1; (G) Experimental demonstration of energy harvesting enabling LED flashing for Design 3-1.

higher surface charge density, as confirmed by experiments: 0.15 mC/m^2 for the non-embedded setup vs. 0.0135 mC/m^2 when embedded in a magnetic elastomer. We optimize the geometry and remanent magnetization distribution of the robots to achieve two on-demand functional modes by dually programming displacement at the control DOFs as well as the output charge on the fixed electrode. The electrodes are assumed to be in a short-circuit configuration. Additionally, we note that pre-compression and/or pre-rotation are applied to facilitate snapping behavior. We impose a volume constraint with an upper bound of 0.3 and a force constraint (constraining the minimal reaction force under a small displacement loading at the control DOFs) to ensure that the optimized design maintains well-connected structural members.

Figure 4D presents the optimized Design 3-1 for a self-powered sensing robot, illustrated in Figure 4A. The robot operates in a working mode (e.g., in contact with tissue) under a downward magnetic field and in a standby mode under an upward magnetic field. The design achieves a pull-push displacement of $\Delta U_{\text{FEA3-1}} =$ 11.2 mm and a corresponding charge output of $\Delta Q_{\text{FEA3-1}} =$ 1078 pC when the magnetic field is flipped from −20 mT to 20 mT. A pre-compression of 1.5 mm is applied symmetrically on both sides to promote snap-through behavior. The large snap displacement enables rapid switching of the two distinct motions, while generating a high output charge for powering carried micro-devices. In the optimized magnetization distribution, horizontally opposed magnetized members are generated on both sides of the organic structure, producing magnetic torques

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that interact with the external field to induce vertical snap bending. A non-magnetized zone is intentionally introduced in the middle of the design to reduce magnetic self-interaction and help with fabrication. The B_a–D response (external magnetic field vs. displacement at the control point), computed using the arc-length method, conforms a characteristic snap-through transition. Finite element simulations of the electric potential distribution (Figure 4D, bottom right) show a clear potential difference between the two actuation states in the air region, confirming the generation of charge output. The output charges on the two electrodes exhibit nearly equal magnitudes (within a numerical error of 0.25%) but opposite flow directions, showing charge conservation between the two deformation modes in the system.

For Design 3-2, we optimize a self-sensing drug-delivery robot that transitions from a closed (drug-encapsulating) to an open (drug-releasing) state under magnetic actuation (Figure 4E), while simultaneously generating an electric signal. The optimized dual-programmable design achieves an open–close displacement of $\Delta U_{\text{FEA3-2}} = 16$ mm and a corresponding charge output of $\Delta Q_{\text{FEA3-2}} = 448$ pC under a magnetic field change from 0 to −20 mT. The generated electric signal can serve as a self-sensing mechanism to monitor the release event, potentially used for feedback-controlled drug delivery. This design exhibits a more complex geometry, as the objective of maximizing charge generation tends to favor vertical pull-push bending (as in Design 3-1), whereas the displacement objective here requires a horizontal open–close motion. These competing requirements call for a more intricate deformation mechanism, which is effectively captured by the inverse-design approach.

In Figure 4F, we fabricate Design 3-1 and conduct experiments to (1) characterize the output charge and compare it with the optimized results, and (2) demonstrate the potential capability of using the electrical waveform to self-sense the motion of the robot. The fixed and compliant electrodes are constructed from an aluminum plate and aluminum foil, respectively, while a PTFE layer functions as the negative charge carrier. The electrodes are connected through a charge amplifier to measure the output charge flowing under short-circuit conditions. Two measurements are conducted corresponding to the robot snapping downward (Experiment 3-1-1) and snapping upward (Experiment 3-1-2). The charge signal drops and jumps within a very short time (less than 0.2 s), and the distinct waveforms can be used to accurately indicate the motion state of the robot. To investigate repeatability, the same test is repeated two more times, showing good consistency: for the snapping downward motion, the absolute output charge $|\Delta Q|$ is 867 pC, 920 pC, and 819 pC, while for the snapping upward motion, $|\Delta Q|$ is 959 pC, 993 pC, and 874 pC. Compared with the FEA-predicted values, the average errors for the two motions are $\bar{e}_{\Delta Q3-1-1} = 24\%$ and $\bar{e}_{\Delta Q3-1-2} = 16\%$, respectively. The error is larger than that of Design Sets 1 and 2 because snapping is a highly transient dynamic process.

In Figure 4G, we demonstrate the same sensing robot, Design 3-1, can also be used to power a small electronic component, such as a white LED. An LED circuit is connected to the electrodes through a bridge rectifier, which converts the alternating current (AC) generated during snap-through into direct current (DC), allowing the LED to flash during both upward and downward snapping events. When the magnetic field is applied using a magnet (instead of a coil) for ease of operation, the robot undergoes snap-through, leading to a rapid change of electric field in the air and redistribution of charges on the electrodes. This transient response induces sufficient AC voltage and current to power the LED. The fabricated sample with pre-compression successfully snaps in both directions and triggers visible LED flashing (also presented in Movie S3). Notably, snapping in the upward direction produces a brighter flash, potentially due to a faster snap-through process resulting from variations in the magnetic field strength generated by the magnet. In the right part of Figure 4G, we also compare with a control sample without pre-compression, which deforms gradually without snap-through and fails to power the LED. These results confirm that snap-through deformation is critical for generating strong electrical output for powering electronics and highlight the potential bio-medical application of dual-programmable magnetic robots capable of energy harvesting and sensing.

3 | Conclusion

We develop a new class of dual-programmable architected magnetic soft materials that enable tuning of magneto–elastic and magneto–electric responses through a multiphysics inverse-design approach, supported by experimental demonstrations. We explore three classes of biomedical applications that leverage this dual-programmable capability. We first study the design optimization of magnetic guidewires to deliver electrical stimulation. A comparison between optimized and intuitive designs shows that the optimized magnetization distribution achieves approximately a 90% improvement, as further confirmed experimentally. We also create wirelessly controlled mechano–electric therapeutic robots capable of delivering both mechanical and electrical stimulation for tissue regeneration. These soft robots can conform to irregular injury sites, illustrating how optimization enables enhanced magneto–electric performance and controlled deformation for patient-specific functionality. In addition, we develop energy-harvesting bio-robots capable of performing functional medical operations, such as sensor deployment and drug release, while also enabling energy harvesting for powering and feedback control. These robots leverage snapping instability, in which magnetically actuated snap-through deformation induces rapid charge redistribution, allowing multifunctional self-powering and self-sensing biomedical systems. An experimental demonstration further shows that the robot can successfully power an LED. Collectively, these optimization-driven designs and experimental validations highlight the translational potential of magnetic soft materials integrated with electrets for next-generation biomedical technologies.

Looking ahead, several considerations remain for advancing this work. For computational modeling, it remains an important future direction to incorporate strongly coupled magneto–mechanical interactions [23, 35, 36] in topology optimization while maintaining satisfactory numerical convergence. From the design perspective, incorporating charge distribution offers an opportunity to broaden the design space. For fabrication, ensuring the long-term durability of embedded charges will be essential for real-world application needs. A promising approach is to introduce nanoparticles in the soft matrix with high

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-voltage poling [8]. Fabrication at smaller scales, for example, magnetic guidewire to be used in blood vessel, require exploration of tailored fabrication approaches, such as electrical discharge machining [37] or direct-ink-writing [2, 38, 39]. For translation to biomedical applications, future designs may consider incorporating tissue modeling to better capture mechanical resistance [26]. A comprehensive electrical characterization, including voltage, current, and power, will provide further insight toward powering micro-devices in practical applications. Integrating suitable bio-safe electrodes will also be an important future direction to enable effective device implantation and ensure long-term stability [40]. For the magnetic guidewire application, it will be important to further explore the interaction with the surrounding environment, including vessel walls, to optimize the guidewire's performance in practical applications. It will also be important to further investigate issues of biocompatibility, encapsulation, and implantation noted in related works [41, 42].

4 | Method and Experimental Section

4.1 | Design Parameterization

The geometry is represented by a physical density field $\bar{\rho}(\mathbf{X})$, mapped from a design variable $\rho(\mathbf{X})$, with $\bar{\rho}(\mathbf{X}) = 1$ for solid and $\bar{\rho}(\mathbf{X}) = 0$ for void (air). The residual magnetic flux density $\mathbf{B}_\mathrm{r}$ is interpolated from the magnetization design variable $\xi^{(k)}(\mathbf{X})$ and optimized among N_m candidate vectors $\mathbf{B}_\mathrm{r}^{(1)}, \ldots, \mathbf{B}_\mathrm{r}^{(N_m)}$, each denoting a distinct direction:

$$\mathbf{B}_\mathrm{r}(\mathbf{X}) = \sum_{j=1}^{N_m} \left(\bar{m}^{(j)}(\mathbf{X})\right)^{p_m} \mathbf{B}_\mathrm{r}^{(j)} \tag{2}$$

where $\bar{m}^{(j)}(\mathbf{X})$ is the physical magnetization variable indicating the j-th remanent magnetization, and p_m is a factor to penalize the mixing of magnetizations.

4.2 | Interpolation of Free Energy

A free energy interpolation scheme based on the energy functions is developed for the solid and void regions. When $\bar{\rho} = 0$, the interpolated energy corresponds to the energy of the void; when $\bar{\rho} = 1$, it corresponds to the energy of the solid. Formally, the interpolated energy function W^{interp} is expressed as

$$\begin{aligned} &W^{\mathrm{interp}}(\mathbf{u}, \varphi; \bar{\rho}, \mathbf{B}_\mathrm{r}) \\ &\quad = (\epsilon + (1-\epsilon)\bar{\rho}^{p_1}) \left(W^{\mathrm{elast}}(\gamma\mathbf{u}) + W^{\mathrm{elast0}}(\mathbf{u}) - W^{\mathrm{elast0}}(\gamma\mathbf{u})\right) \\ &\qquad + \bar{\rho}^{p_2} W^{\mathrm{mag}}(\mathbf{u}, \mathbf{B}_\mathrm{r}) + \bar{\rho}^{p_3} W^{\mathrm{elec}}(\mathbf{u}, \varphi) + (1 - \bar{\rho}^{p_3}) W^{\mathrm{elec0}}(\mathbf{u}, \varphi) \end{aligned} \tag{3}$$

where the displacement vector $\mathbf{u}$ is related to the deformation gradient $\mathbf{F}$ through $\mathbf{F} = \mathbf{I} + \nabla\mathbf{u}$, and φ denotes the electric potential defined by the electric field $\mathbf{E} = -\nabla\varphi$. The parameter ϵ is a small value introduced to avoid numerical singularities, while p_1, p_2, and p_3 represent SIMP (solid isotropic material with penalization [43])-like penalization factors for the elastic W^{elast}, magnetic W^{mag}, and electric W^{elec} energies, respectively (where the subscript 0 indicates these energies for the air region. The mechanical energy follows the filled elastomer model from [44]. To improve numerical stability under large deformations during the optimization process, we adopt an energy interpolation scheme between hyperelastic (W^{elast}) and linear elastic (W^{elast0}) energies, with γ as the interpolation factor on displacement [45]. The magnetic energy (W^{mag}) is taken from a decoupled hard-magnetic soft material model [22], where the contribution from the air domain is neglected for simplification and a uniform magnetic field is assumed. The electric energy follows the formulation in [7], interpolated between the values corresponding to the material (W^{elec}) and to the air region (W^{elec0}). The detailed expressions are presented in Section S1.

4.3 | Magneto–Mechano–Electric Governing Equations

The coupled-field governing equations are formulated with interpolated energy and solved through the balance of linear momentum together with Maxwell's equations, as given below:

$$\begin{cases} \nabla \cdot \dfrac{\partial W^{\mathrm{interp}}}{\partial \mathbf{F}} + \mathbf{f}_{\mathrm{b0}} = \mathbf{0} \quad \text{in} \quad \mathcal{D}_0, \\ \nabla \cdot \dfrac{\partial W^{\mathrm{interp}}}{\partial \mathbf{E}} - \rho_{\mathrm{q0}} = \mathbf{0} \quad \text{in} \quad \mathcal{D}_0, \\ \mathbf{u} = \tilde{\mathbf{u}} \quad \text{on} \quad \Gamma_\mathrm{u}, \\ \dfrac{\partial W^{\mathrm{interp}}}{\partial \mathbf{F}} \cdot \mathbf{N} = \tilde{\mathbf{t}} \quad \text{on} \quad \Gamma_\mathrm{t}, \\ \varphi = \tilde{\varphi} \quad \text{on} \quad \Gamma_\varphi, \\ -\dfrac{\partial W^{\mathrm{interp}}}{\partial \mathbf{E}} \cdot \mathbf{N} = \tilde{\omega} \quad \text{on} \quad \Gamma_\omega \end{cases} \tag{4}$$

The solid domain and the surrounding air under the reference configuration are defined by $\mathcal{D}_0$. A very soft material model is used to represent the air in the optimization. The domain is subjected to an applied displacement field $\tilde{\mathbf{u}}$ on the boundary Γ_u and an applied traction $\tilde{\mathbf{t}}$ on Γ_t, where $\mathbf{N}$ is the unit normal to the boundary. The electric potential field $\tilde{\varphi}$ is applied on $\Gamma_{\tilde{\varphi}}$, and the electric boundary charge per unit area $\tilde{\omega}$ is applied on Γ_ω. These boundaries satisfy $\Gamma_\mathrm{u} \cup \Gamma_\mathrm{t} = \partial\mathcal{D}_0$, $\Gamma_\mathrm{u} \cap \Gamma_\mathrm{t} = \emptyset$, $\Gamma_\varphi \cup \Gamma_\omega = \partial\mathcal{D}_0$, and $\Gamma_\varphi \cap \Gamma_\omega = \emptyset$.

4.4 | Optimization Formulation

The proposed formulation enables dual programming of electrical output and mechanical deformation via joint optimization of geometry and remanent magnetization. A total of N_ℓ load cases under different boundary conditions and external magnetic fields are considered, denoted as $\mathbf{B}_\mathrm{a}^{(\ell)}$. The FEA solutions are $(\mathbf{u}^{(\ell)}, \varphi^{(\ell)})$ for $\ell = 1, 2, \ldots, N_\ell$. A dual-objective function for programming both magneto-elastic and magneto–electric responses (with $N_\ell = 2$ to optimize the performance from two quasi-static

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configurations) is formulated:

$$f = \min_{i,j}\left(w_{\mathrm{Q}} f_{\mathrm{Q}}^{(i)}, w_{\mathrm{U}} f_{\mathrm{U}}^{(j)}\right) \quad \text{with} \quad \begin{cases} f_{\mathrm{Q}}^{(i)} = C_{\mathrm{Q}} \dfrac{\Delta Q^{(i)}}{A^{(i)} \rho_{\mathrm{q0}}}, & i = 1, \ldots, N_{\mathrm{Q}} \\ f_{\mathrm{U}}^{(j)} = C_{\mathrm{U}} \dfrac{\Delta U^{(j)}}{L_{\mathrm{c}}}, & j = 1, \ldots, N_{\mathrm{U}} \end{cases} \tag{5}$$

where w_{Q} and w_{U} are weighting factors of choice. The indices i and j correspond to the i-th grounded or short-circuited electrode surface ($\varphi = 0$) and the j-th control DOF for displacement, respectively. The symbol $A^{(i)}$ represents the area of the i-th electrode, ρ_{q0} is the embedded charge density, and L_{c} is the characteristic length of the design domain. The constants C_{Q} and C_{U} are incorporated in the objective function to regularize the two objective values, ensuring they are comparable to avoid numerical singularity. We employ a min-max formulation [46] for the multi-objective function to ensure the objective values are well-balanced.

The output charge $\Delta Q^{(i)}$ over the surface area $A^{(i)}$ of the i-th electrode [7] is given by

$$\frac{\Delta Q^{(i)}}{A^{(i)}} = \frac{1}{A^{(i)}} \int_{\Gamma_{\mathrm{Q}}^{(i)}} \left(\mathbf{D}^{(2)} \cdot \mathbf{N} - \mathbf{D}^{(1)} \cdot \mathbf{N}\right) \mathrm{d}S \tag{6}$$

where $\mathbf{D}^{(2)}$ and $\mathbf{D}^{(1)}$ represent the electric flux density under the applied magnetic fields $\mathbf{B}_{\mathrm{a}}^{(2)}$ and $\mathbf{B}_{\mathrm{a}}^{(1)}$, respectively. The integration is performed over the surface of $\Gamma_{\mathrm{Q}}^{(i)}$.

The displacement change at the j-th control DOF is given by

$$\Delta U^{(j)} = \mathbf{u}^{(2)}(\mathbf{X}^{(j)}) \cdot \mathbf{e} - \mathbf{u}^{(1)}(\mathbf{X}^{(j)}) \cdot \mathbf{e} \tag{7}$$

where $\mathbf{u}^{(2)}(\mathbf{X}^{(j)})$ and $\mathbf{u}^{(1)}(\mathbf{X}^{(j)})$ represent the displacements at the location $\mathbf{X}^{(j)}$ under the applied magnetic fields $\mathbf{B}_{\mathrm{a}}^{(2)}$ and $\mathbf{B}_{\mathrm{a}}^{(1)}$, respectively. The vector $\mathbf{e}$ is a unit directional vector, with its direction depending on the specific optimization problem.

With the objective function defined, the optimization problem is formulated as follows:

$$\begin{cases} \underset{0\le\rho\le1, 0\le\xi^{(k)}\le1}{\text{maximize}} & f(\bar{\rho}(\rho), \mathbf{B}_{\mathrm{r}}(\xi^{(k)}), \mathbf{u}^{(\ell)}(\bar{\rho}, \mathbf{B}_{\mathrm{r}}), \phi^{(\ell)}(\bar{\rho}, \mathbf{B}_{\mathrm{r}})), \\ & k = 1, 2, \cdots, N_m, \quad \ell \in 1, 2, \cdots, N_\ell \\ \text{subject to} & \begin{cases} G_{\mathrm{V}}(\bar{\rho}) := |\Omega_0|^{-1} \int_{\Omega_0} \bar{\rho} \mathrm{d}\mathbf{X} - V_{\max} \le 0, \\ G_{\mathrm{F}}(\mathbf{u}^{(\ell)}, \phi^{(\ell)}) := \mathbf{F}_{\mathrm{int}}(\mathbf{u}^{(\ell)}, \phi^{(\ell)}) \cdot \mathbf{L} - F_{\min} \ge 0, \\ (\mathbf{u}^{(\ell)}, \phi^{(\ell)}) \text{ solution of the boundary value problem} \end{cases} \end{cases} \tag{8}$$

In the optimization formulation, a volume constraint G_{V} is imposed on the physical density variable $\bar{\rho}$ to limit the overall volume usage, with an upper bound of $V_{\max}$. The constraint function G_{F} ensures that the global internal force in the region of interest, computed as the inner product of the internal force vector $\mathbf{F}_{\mathrm{int}}$ and the prescribed output force indicator $\mathbf{L}$, exceeds a predefined lower bound $F_{\min}$. The indicator vector $\mathbf{L}$ specifies the DOFs of interest, where $L_i = 1$ denotes a controlled DOF and $L_i = 0$ otherwise.

4.5 | Numerical Implementation

The optimization problem is solved by the method of moving asymptotes (MMA)[47]. The nested coupled magneto–electro–mechanical problem solved using finite element method. The nonlinear system of equations is solved using the Newton–Raphson method. Both inverse-design optimization and finite element analyzes are implemented using the FEniCSx platform [48, 49]. Details of the computational mode are provided in Section S1. We note that, in the optimization, the electro-induced stress and far-field air are neglected for numerical stability, with justification and comparative analysis provided in Section S3.

4.6 | Fabrication Protocol

The fabrication procedure involves two main stages: manufacturing magnetic elastomers and embedding static charge, as illustrated in Figure 5A. The hard-magnetic elastomer is synthesized using Ecoflex 00-30 (Smooth-On, Inc.) mixed with Nd-Pr-Fe-B magnetic particles (20173, Magnequench) with a median diameter of 5 μm. The mass ratio of Ecoflex to magnetic particles is 4:5. Ecoflex Part A, Part B, and magnetic particles are mixed in a planetary mixer (AR-100, Thinky) at 1400 rpm for 90 s to ensure thorough mixing and defoaming. The prepared mixture is poured or injected into polylactic acid (PLA) molds, which are fabricated using a fused deposition modeling (FDM) 3D printer (Original Prusa i3 MK3S+) according to optimized design geometries. After being cured at room temperature for at least 24 h, the parts are demolded and individually magnetized based on the optimized magnetization directions using a 2 T magnetic field generated by an impulse magnetizer (IM-10-30, ASC Scientific). The components are bonded using silicone rubber adhesive (Sil-Poxy, Smooth-On, Inc.).

To fabricate ME guidewires and therapy robots from magnetic elastomers, we apply electrodes with carbon conductive grease (846, MG Chemicals) and ground them to maintain charge balance and ensure electrostatic stability throughout the charge embedding process. A 25 μm polytetrafluoroethylene (PTFE) thin film (McMaster–Carr) serves as the charge carrier and is placed onto magnetic elastomer with the bottom grounded. The PTFE film is divided into smaller pieces to minimize its influence on the mechanical properties of the elastomer while leaving small regions for adhesive bonding. PTFE is triboelectrically charged by rubbing it with wool felt (McMaster–Carr), which easily retains positive charges, while PTFE demonstrates a strong ability to carry negative charges [50]. The pieces are carefully bonded with silicone rubber adhesive. The assembled sample is then mounted onto a customized 3D-printed support through clamping or bonding according to the required mechanical boundary conditions. The sample is treated with anti-static spray (McMaster–Carr) to neutralize surface charges in regions not covered by the electrode.

To fabricate the energy-harvesting functional bio-robots, the PTFE layer is directly placed on an aluminum plate, which serves as the bottom electrode, and triboelectricity charged by rubbing the surface. The cast magnetic elastomer is mounted under pre-compression on a PLA fixture. An aluminum foil is attached to the appropriate surface, functioning as the top electrode. The assembled fixture is then placed on top of the charged PTFE layer.

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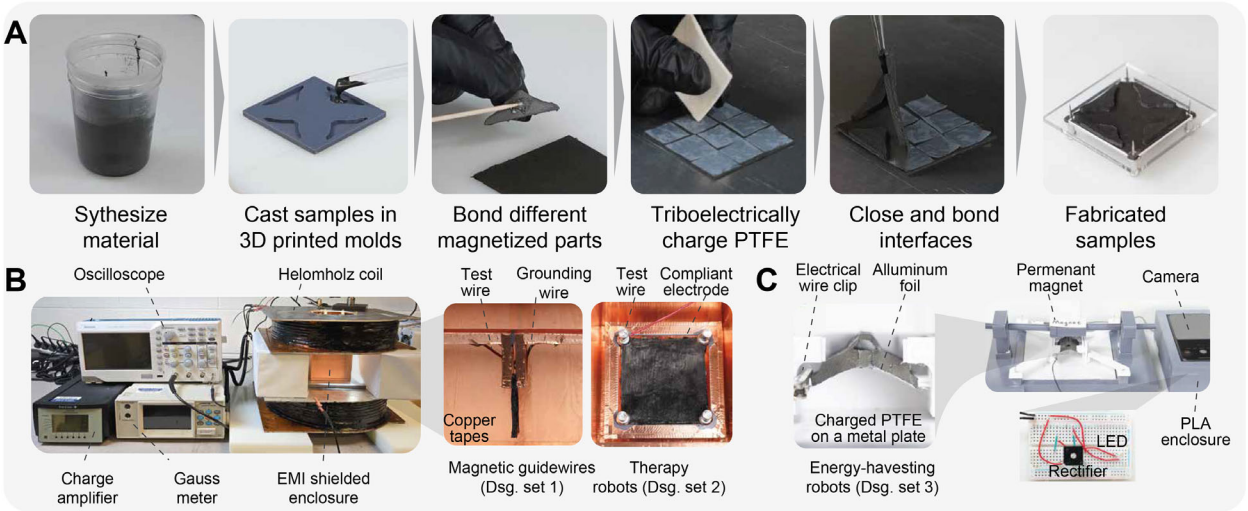


FIGURE 5 | Hybrid fabrication and experimental setups: (A) Fabrication process using mold-casting and triboelectric charge embedding for Design 2-2; (B) Experimental setup for measuring output charge under magnetic actuation for magnetic guidewires and therapy robots; (C) Experimental setup showing LED powered by an energy-harvesting functional robot.

4.7 | Experimental Protocol

The experimental setup consists of three main components: the actuation system, the sample enclosure and fixture, and the measurement apparatus, as shown in Figure 5B. ForDesigns 1-1, 1-2, and 2-2, a nearly uniform magnetic field is applied using a customized Helmholtz coil (Woodruff Scientific, Inc.). To generate the unidirectional field, a cyclic sinusoidal DC current is supplied to the coil through a programmable power supply (ADG-L, Preen). A Gauss meter (MFM 4000, PCE) is used to measure the applied magnetic field, allowing fine-tuning of the electric current to achieve the target field strength. For Design 3-1, a rectangular permanent magnet mounted on a PLA frame is used to actuate the magnetic material and induce snap-through behavior.

For the sample enclosure, we aim to minimize electromagnetic interference (EMI) by using an EMI-shielded enclosure. This custom-built enclosure is made with copper tape applied to a PLA structure, printed using FDM printing. To prevent interference from surface charges on the PLA, both the interior and exterior of the PLA are covered with conductive copper tape, ensuring that the PLA does not carry charges that could disrupt experimental measurements. The enclosure is grounded via a grounding cable. The sample fixtures are designed and also printed in PLA, with similar copper tape treatment applied. When handling and installing the fabricated sample onto the fixture, we use a grounded metal tweezer to ensure that the electrostatic stability of the sample is maintained.

For the deformation measurement, we use a high-resolution camera (α7R, SONY) to capture the sample deformation during the experiment. We use a depth caliper to measure the out-of-plane deformation of Design 2-2. For the output charge measurement, a grounded charge amplifier (Nexus 2692, Brüel & Kjær) amplifies the charge signals. A customized TNC cable carries the input signal to the amplifier. The charge amplifier outputs voltage signals, which are recorded by an oscilloscope (TBS1052C, Tektronix). To minimize EMI (particularly from the 60 Hz power line frequency in the U.S.), we filter out high-frequency components above 20 Hz. For the LED powering experiment indicated in Figure 5C, we construct a rectifier-integrated LED circuit, in which the rectifier is connected to the two electrodes. To enhance the visibility of LED flashing for Design 3-1, the LED circuit is placed inside a dark enclosure. A camera records the flashes through a small viewing hole. A multimeter is used to verify circuit conductivity.

4.8 | Material Characterization

We conduct comprehensive electrical, mechanical, and magnetic measurements for characterizing material properties. The surface charge density, ρ_q, is a critical parameter governing the output charge performance. As shown in Figure 2G, the surface charge density is characterized using an ideal parallel-plate capacitor setup [6, 51] by a non-contact voltmeter (Trek 884). The test sample consisted of a cylindrical hard-magnetic elastomer (the out-of-plane thickness is 1.1 mm) with a PTFE thin film (the out-of-plane thickness is 25 μm) attached to its top surface and a compliant electrode on the bottom. The sample is placed on a grounded metal plate with the electrode in contact. The PTFE surface is triboelectrically charged by controlled and consistent rubbing for all samples, including those fabricated for the optimized designs. The surface voltage V_s is measured using the non-contact voltmeter at a probe-to-sample distance of 45 mm. The surface charge density is calculated as

$$\rho_{q0} = \frac{\varepsilon_{rs}\varepsilon_0 V_s}{H_s} \tag{9}$$

where H_s is the sample thickness, ε_{rs} is the relative permittivity (measured by dielectric spectrometer), and ε_0 is the vacuum permittivity. We took multiple measurements at different spots for three sample replicates.

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For Designs 1-1 and 1-2, two charged PTFE layers are sandwiched between two grounded magnetic elastomers; the effective charge density is taken as $\rho_{q0} = -0.027\ \mathrm{mC/m^2}$. For Design 2-2, only one charged PTFE layer with a magnetic elastomer is used, resulting in $\rho_{q0} = -0.0135\ \mathrm{mC/m^2}$. For Design 3-2, the PTFE film is directly placed on a grounded metal plate to enlarge embedded charges and measured following the same procedure, but without the magnetic elastomer, yielding $\rho_{q0} = -0.15\ \mathrm{mC/m^2}$.

For the other material properties, uniaxial tension–compression tests are performed to determine the hyperelastic constants, giving $\alpha_1 = 0.6497$, $\mu_1 = 0.0116$ MPa, $\alpha_2 = 0.8307$, and $\mu_2 = 0.0277$ MPa. The magnetic particle mass density is measured using a pycnometer, yielding $\rho = 0.172\ \mathrm{g/cm^3}$. Based on the measured density of magnetic elastomer, the magnitude of the body force density is calculated as $|\mathbf{f}_{b0}| = 1.83\ \mathrm{N/cm^3}$. The relative permittivity is measured using a dielectric spectrometer (Novocontrol Concept 47 Analyzer), giving $\varepsilon_r = 6.9$. The residual magnetic flux density is measured using a vibrating-sample magnetometer (Quantum Design MPMS3), yielding $\mathbf{B}_r = 106$ mT. Details of these characterization experiments are provided in Section S2. For biocompatibility, the same type of magnetic material has been tested and shown to be biocompatible in [26], and the PTFE material used for the electret has been proven biocompatible in multiple in vivo studies [52].

4.9 | Statistical Analysis

The results for material property characterization are reported as mean ± standard deviation in Figure 2G, with a sample size of $n = 3$. MATLAB is used for data analysis and statistical analysis.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: adfm75161-sup-0001-SuppMat.pdf.

Supporting File 2: adfm75161-sup-0002-MovieS1.mp4.

Supporting File 3: adfm75161-sup-0003-MovieS2.mp4.

Supporting File 4: adfm75161-sup-0004-MovieS3.mp4.

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